

Neurogenic Hypertension, Including Hypertension Associated With Stroke or Spinal Cord Injury

Venkatesh Aiyagari, Philip B. Gorelick

An intimate relationship exists between the nervous system and blood pressure (BP).¹ It is well recognized that the elevated BP response to stressors is mediated by the sympathetic nervous system (SNS). However, the role of the SNS in long-term regulation of BP and the initiation and maintenance of hypertension is now better delineated. Several studies of serum catecholamine levels, renal norepinephrine spillover, microneurography, and heart rate variability suggest that sympathetic activation plays a major role in hypertensive patients.² The SNS also has an important role in hypertension after neurologic injury. This chapter discusses the physiology and management of hypertension in such injury.

PHYSIOLOGY AND PATHOPHYSIOLOGY

Neural Control of Blood Pressure

The brainstem, especially the ventral medulla, has a key role in the maintenance of BP (Fig. 41.1). BP is controlled by the nucleus tractus solitarius, which receives inhibitory baroreceptor afferents, and the rostral ventrolateral medulla and rostral ventromedial medulla, which are the source of excitatory descending bulbospinal pressor pathways. In addition, a depressor center in the caudal ventrolateral medulla composed of γ -aminobutyric acid (GABA)-containing neurons receives afferents from the nucleus tractus solitarius and projects to the rostral ventral medulla. These inhibitory GABA-containing neurons are tonically active, and reduced activity of these neurons leads to hypertension.^{3–5}

The ultimate effector units are the sympathetic neurons located in the intermediolateral cell column of the spinal cord and the parasympathetic neurons found in the dorsal motor nucleus of the vagus and nucleus ambiguus located in the medulla. In addition, impulses from the limbic system, cerebral cortex, and hypothalamus directly or indirectly project to the intermediolateral cell column of the spinal cord and influence BP regulation.

The factors that lead to increased sympathetic activation in hypertension are poorly understood. However, there is strong evidence linking hypertension with increased levels of circulating inflammatory markers, such as tumor necrosis factor- α , interleukin-6, C-reactive protein, monocyte chemoattractant protein 1, and adhesion molecules such as P-selectin and intercellular adhesion molecule 1. Angiotensin II (Ang II) and aldosterone also play a crucial role in vascular inflammation, and treatment with both candesartan and mineralocorticoid antagonists decrease the levels of inflammatory markers. In addition, Ang II-mediated hypertension is associated with brain microglial activation and increased brain levels of inflammatory cytokines and reactive oxygen species. An increase in reactive oxygen species may directly activate or sensitize sympathetic neurons and scavenge nitric oxide, the latter of which tonically inhibits sympathetic outflow. Thus,

a dysfunction of the neural-immune-vascular triad leading to an increase in central oxidative stress may drive sympathetic activation, which then increases Ang II and promotes further inflammation and vascular dysfunction.⁶ Another matter for debate is whether the kidney or the central nervous system (CNS) is the main driver for increased sympathetic tone in patients with neurogenic hypertension. The role of the afferent and efferent renal nerves in mediating neurogenic hypertension is being researched, and renal denervation procedures are being explored as possible therapeutic options to treat hypertension.

Cerebrovascular Autoregulation

Under normal conditions, cerebral blood flow (CBF) of the adult brain is approximately 50 mL/100 g/min. CBF is regulated by the relationship between cerebral perfusion pressure (CPP) and cerebrovascular resistance (CVR):

$$CBF = CPP/CVR$$

CPP is the difference between the mean arterial blood pressure (MAP) and the intracranial pressure (ICP). If ICP is increased, systemic BP must also increase to maintain CPP and CBF.

Cerebrovascular autoregulation maintains a constant blood flow over a wide range of CPP. Normally, changes in CPP have little effect on CBF because of compensatory changes in CVR. An increase in CPP produces vasoconstriction and a decrease produces vasodilation, thus keeping the CBF constant (Fig. 41.2). Autoregulation is effective for a range of CPP from about 60 to 150 mm Hg. In chronically hypertensive individuals, the cerebral arterioles develop medial hypertrophy and lose the ability to dilate effectively at lower pressures. This leads to a shift of the autoregulatory curve to the right.⁷ In these individuals, a rapid reduction of BP may lead to a drop in CBF even though the BP might still be “normal.” With effective control of hypertension for several months, the normal range for autoregulation can be reestablished.⁸

Above the upper limit of autoregulation, there is breakthrough vasodilation leading to damage of the blood-brain barrier, cerebral edema, and possibly cerebral hemorrhage. Below the lower limit of autoregulation, decreases in CPP lead to a decrease in CBF. Under these circumstances, increased extraction of oxygen and glucose maintains normal cerebral metabolism and brain function. When the CBF decreases to less than 20 mL/100 g/min, increases in oxygen extraction are no longer able to supply the metabolic needs of the brain, leading to impairment of brain function.

NEUROGENIC HYPERTENSION

It is challenging to identify patients in whom neurogenic mechanisms are the main driver of hypertension. There are no readily available

Neural Pathways Involved in the Control of Blood Pressure

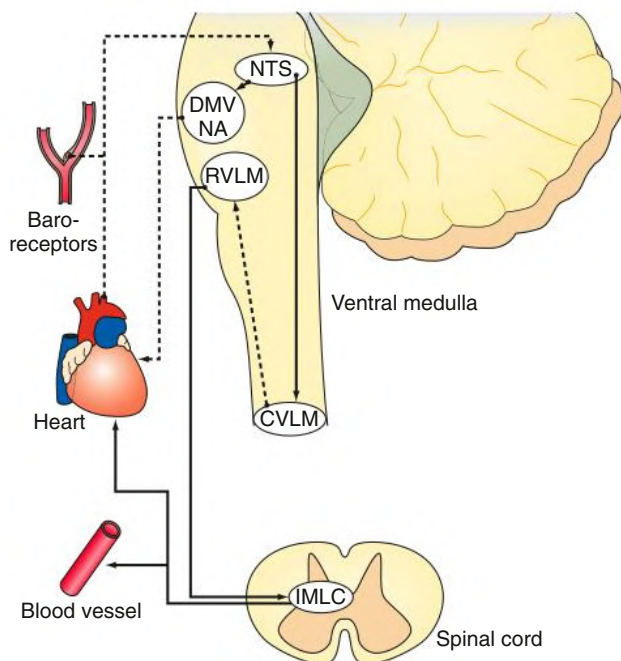


Fig. 41.1 Neural Pathways Involved in the Control of Blood Pressure. The ventral medulla has a key role in generating both excitatory (solid line) and inhibitory (dotted line) pathways, largely through the rostral ventrolateral medullary neurons (RVLM) and nucleus tractus solitarius (NTS), respectively. Ultimate effector control is provided by sympathetic activation originating in the intermediolateral cell column (IMLC) and parasympathetic action through the nucleus ambiguus (NA) and dorsal motor nucleus of the vagus nerve (DMV). CVLM, Caudal ventrolateral medullary neurons.

clinical tests to evaluate increased sympathetic activity and differentiate neurogenic hypertension from nonneurogenic hypertension. Nonetheless, some clinical clues may suggest the presence of neurogenic hypertension. These include paroxysmal or labile hypertension, young age of onset, severe hypertension that is resistant to angiotensin-converting enzyme (ACE) or diuretic therapy and hypertension associated with ingestion of sympathomimetic agent.⁹

SPECIFIC SYNDROMES

Hypertension After Stroke

Epidemiology

BP is commonly elevated in patients with stroke. The reason for this acute hypertensive response after stroke remains a matter of conjecture. In a large retrospective analysis that examined 276,734 patients presenting to the emergency department (ED) with acute ischemic stroke, the incidence of elevated BP was 76.5%.¹⁰ The Chinese Acute Stroke Trial (CAST) and the International Stroke Trial (IST), each enrolling approximately 20,000 patients with ischemic stroke, reported systolic BP (SBP) above 140 mm Hg in 75% and 80% of patients and severely elevated SBP of greater than 180 mm Hg in 25% and 28%, respectively.^{11,12}

Hypertension is the most important modifiable risk factor for stroke, and reduction in BP is effective in the primary prevention of both ischemic and hemorrhagic stroke; BP reduction also decreases the risk for a recurrent ischemic or hemorrhagic stroke.¹³ Combined

Cerebral Autoregulation Curve

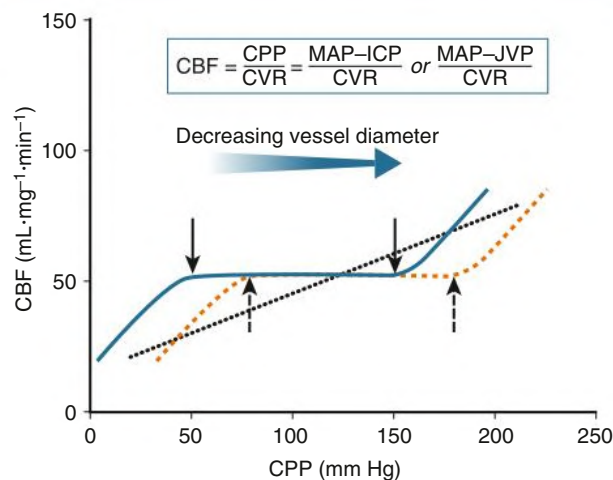


Fig. 41.2 Cerebral Autoregulation Curve. In the normal state (solid line), the cerebral blood flow (CBF) is held constant across a wide range of cerebral perfusion pressure (CPP) (60–150 mm Hg). In chronic hypertension (dashed line), the autoregulation curve shifts to the right. In the presence of acute cerebral ischemia (dotted line), cerebral autoregulation may be impaired, and the CBF becomes dependent on the CPP. CVR, Cerebral vascular resistance; ICP, intracranial pressure; JVP, jugular venous pressure; MAP, mean arterial pressure. (From Testai FD, Aiyagari V. Acute hemorrhagic stroke pathophysiology and medical interventions: blood pressure control, management of anticoagulant-associated brain hemorrhage and general management principles. *Neurol Clin.* 2008;26:963–985.)

data from 40 trials show that a 10% reduction in SBP lowers stroke risk by one-third.¹⁴ A 5-mm reduction in diastolic pressure together with a 9-mm lower SBP confers a 33% lower risk for stroke, and a 10-mm lower diastolic BP (DBP) together with an 18- to 19-mm lower SBP confers more than a 50% reduction in stroke risk.¹⁵ In patients who have had a stroke, the Perindopril Protection Against Recurrent Stroke Study (PROGRESS) showed that BP reduction was associated with reductions of 28% in stroke recurrence and 26% in major coronary and vascular events, even in normotensive subjects. This study also demonstrated a reduction of absolute rates of hemorrhagic stroke from 2% to 1% over a mean of 3.9 years of follow-up.¹⁶

However, the management of BP in the immediate aftermath of a stroke is controversial.¹⁷ A high proportion of patients have elevated BP immediately after a stroke, but BP spontaneously decreases over 1 to 2 weeks to the prestroke baseline in most patients. Some of the postulated causes of elevated BP are listed in Box 41.1. An increased BP after stroke is associated with a higher mortality. Nonetheless, it is uncertain whether this increase directly contributes to poor outcome and whether immediate lowering of BP will lead to better outcomes. SBP variability may be associated with the risk for cerebrovascular events independent of the mean BP. Home BP monitoring and ambulatory BP monitoring are useful in monitoring BP variability. Normally, BP is highest in the morning and gradually falls to reach its lowest level during sleep. Several alterations in this pattern, including a lack of nighttime decline in BP (nondippers), a rise in nighttime BP (reverse dippers), and a more than 20% fall in nighttime BP (extreme dippers), are associated with increased risk for vascular complications, including stroke. These findings also have a bearing on the choice of the antihypertensive agent for stroke prevention. Calcium channel blockers (CCBs) and nonloop diuretics appear to decrease SBP variability,

BOX 41.1 Postulated Causes of Hypertension After Stroke

- Preexisting hypertension
- “White coat” effect
- Stress of hospitalization
- Cushing reflex^a
- Catecholamine and cortisol release
- Lesion of brainstem or hypothalamus
- Nonspecific response to brain damage

^aHypothalamic response to raised intracranial pressure or ischemia consisting of hypertension with bradycardia.

whereas β -blockers, angiotensin receptor blockers (ARBs), and ACE inhibitors do not appear to decrease BP variability.

Pathophysiology

An understanding of cerebrovascular pathophysiology is essential to understand the pros and cons of treating hypertension in these patients (Table 41.1).

In patients with an ischemic stroke, vascular occlusion leads to a central region of irreversibly ischemic brain surrounded by an ischemic zone where blood flow is reduced but brain tissue is still viable (the so-called penumbra zone). After 2 or 3 days, the ischemic areas either recover or undergo infarction. In the first few days, perfusion of the area surrounding that destined to infarction is marginal and a further decrease in blood flow might lead to infarction there as well. Because cerebral autoregulation is impaired with acute ischemic stroke, a fall in BP could lower blood flow and extend infarction. On the other hand, a very high BP could exacerbate cerebral edema or lead to hemorrhagic transformation, especially if thrombolytic agents have been given.

In patients with intracerebral hemorrhage (ICH), the considerations are different.¹⁸ Hematoma expansion occurs in 73% of patients within 24 hours, and significant expansion (>33% increase in volume) occurs in nearly 40% of patients.¹⁹ Hematoma expansion is frequently associated with decline in neurologic status and is an independent predictor of mortality and poor functional outcome. Therefore, BP is often lowered in these patients in the hope that this might decrease hematoma expansion. However, it is not often clear if the elevation in BP is the cause or the consequence of hematoma expansion. The suggestion that there may be perihematomal ischemia around an ICH has not been supported by recent studies.²⁰ Furthermore, some patients with ICH might have increased ICP because of the hematoma volume or associated hydrocephalus. In such a situation, lowering of BP is not warranted because it might critically lower CPP. Monitoring of ICP and CPP may be helpful in choosing the appropriate BP target in these circumstances.

In patients with aneurysmal subarachnoid hemorrhage (SAH), there is a significant risk for rebleeding from aneurysmal rerupture, and therefore early BP control is recommended to decrease the rebleeding risk. Some patients with SAH have associated myocardial dysfunction (stunned myocardium), in which case high BP might worsen myocardial function. Like patients with ICH, in patients with hydrocephalus or an associated ICH, ICP and CPP monitoring can help guide BP management. However, many patients develop vasospasm of the intracranial arteries at 4 to 12 days after SAH, and reduction of BP may lead to worsening of cerebral ischemia in this situation. Therefore, once the aneurysmal rupture has been adequately treated with surgical clipping or coiling, BP is usually maintained at a normal or slightly elevated level in these patients.

TABLE 41.1 Acute Treatment of Hypertension in Stroke: Advantages and Disadvantages

Advantages	Disadvantages
Acute Ischemic Stroke	
Might lower mortality	BP decreases on its own
Might decrease stroke progression	No proven benefit
Might decrease hemorrhagic transformation (especially after tPA)	Ongoing ischemia around infarct (ischemic penumbra)
Might decrease cerebral edema formation	Altered autoregulation from chronic hypertension, ischemia
Might be helpful for systemic reasons (e.g., associated myocardial ischemia)	Large-vessel stenosis might have resulted in reduction of perfusion
Patients likely to be more compliant with antihypertensive use if treatment initiated in hospital	Chance of propagating thrombus
	Anecdotal case reports and trial results demonstrating deterioration with BP decrease
	Principle of “do no harm” (primum non nocere)
Acute Intracerebral Hemorrhage	
Might lower mortality	BP decreases on its own
Might decrease hematoma expansion	No proven benefit
Might decrease cerebral edema formation	Possible zone of ischemia around intracerebral hematoma
Might be helpful for systemic reasons (e.g., associated myocardial ischemia)	Chronically hypertensive patients require higher CPP because of shift in autoregulatory curve
Patients likely to be more compliant with antihypertensive use if treatment initiated in hospital	ICP may be elevated, and lowering BP reduces what could be marginal CPP
	Principle of “do no harm” (primum non nocere)
Aneurysmal Subarachnoid Hemorrhage	
Might decrease rebleeding rate	No proven benefit
Might help if there is cardiac ischemia (stunned myocardium)	ICP may be elevated, and lowering BP reduces what could be marginal CPP
	Might lead to cerebral ischemia in presence of vasospasm

BP, Blood pressure; CPP, cerebral perfusion pressure; ICP, intracranial pressure; tPA, tissue plasminogen activator.

Diagnosis and Treatment

Acute management of hypertension in stroke is highly dependent on the type of stroke (ischemic vs. hemorrhagic) and, for ischemic stroke, the use of thrombolysis. The benefits of lowering BP to prevent hematoma expansion in the setting of ICH or hemorrhagic transformations of ischemic stroke should be balanced with the risk for abrupt reduction of CBF in chronically hypertensive patients with shifted cerebrovascular autoregulation, especially if increased ICP and cerebral edema are present.

It is important to distinguish between hypertensive encephalopathy, in which lowering of BP is clearly indicated, and ischemic stroke with hypertension, in which urgent lowering of BP may not be warranted. The level of consciousness, the presence of focal neurologic deficits, and the funduscopic examination can help in making this distinction. Hypertensive encephalopathy is a syndrome of global neurologic dysfunction, usually with papilledema. Focal neurologic deficits are usually less prominent. In acute ischemic stroke, the focal neurologic deficit is prominent, and the symptoms and neurologic signs

Management of Hypertension Following Acute Stroke

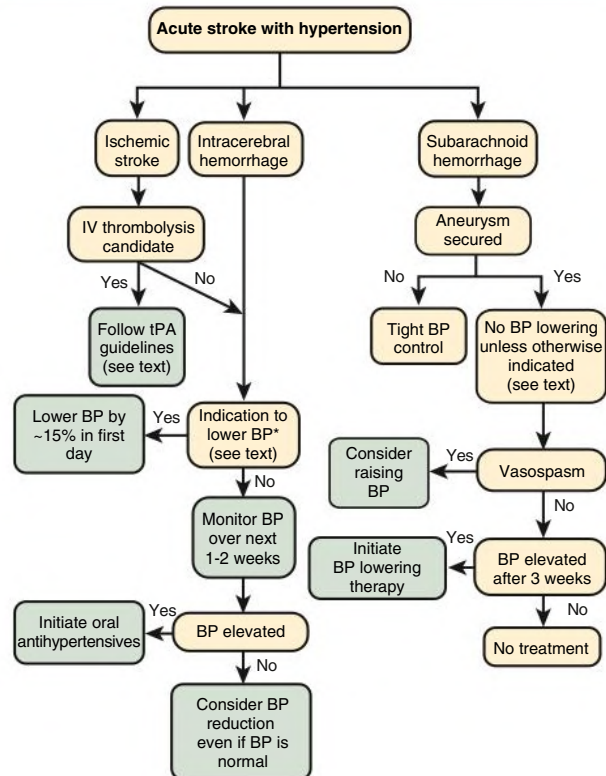


Fig. 41.3 Management of Hypertension After Acute Stroke. *Indication for treatment includes systolic blood pressure (BP) >220 mm Hg or diastolic BP >120 mm Hg for ischemic stroke, the presence of associated conditions such as aortic dissection or myocardial infarction, and, in cases of cerebral hemorrhage, systolic BP >180 mm Hg or mean arterial pressure >130 mm Hg. IV, Intravenous; tPA, tissue plasminogen activator.

often can be mapped to the vascular territory supplied by a specific cerebral blood vessel. Early alterations of consciousness are less common, except with brainstem strokes or when there is “malignant” brain edema secondary to a massive hemispheric infarction.

Several recent studies assess the benefits and risks of BP lowering after ischemic and hemorrhagic stroke. A Cochrane review on this topic concluded that more research is needed to evaluate and identify the candidates most likely to benefit from management of BP in the acute phase of stroke and the timeframe for such intervention.²¹

An overview of recommendations for treating BP in different clinical situations is outlined in Fig. 41.3.

Acute ischemic stroke. Several large prospective trials of BP lowering in acute ischemic stroke have been completed and are summarized in Table 41.2.^{22–32}

The current guidelines of the American Heart Association/American Stroke Association (AHA/ASA) and the European Stroke Organisation (ESO) for BP management in acute ischemic stroke recommend a cautious approach to lowering BP in acute ischemic stroke. Patients receiving thrombolytic therapy should have their BP lowered to less than 185/110 mm Hg before the administration of the thrombolytic agent and maintained at a level of less than 180/105 mm Hg for at least the first 24 hours after thrombolytic treatment. Patients not

receiving thrombolytic agents should have antihypertensive medications withheld unless the SBP is greater than 220 mm Hg or the DBP is greater than 120 mm Hg, in which case a 15% reduction in BP in the first 24 hours appears reasonable. Recommendations are summarized in Table 41.3.^{33,34}

Thus, available evidence does not support immediate BP reduction after an acute ischemic stroke. Based on the limited data and natural history of BP after ischemic stroke, treatment of newly diagnosed previously untreated hypertension or resumption of long-term antihypertensive medication can be initiated or resumed gradually after the first 24 hours. Reduced dosage and/or number of agents from the prestroke regimen should be used to avoid rapid reduction of BP in the case of prior outpatient noncompliance. It may be advisable to administer BP-lowering agents when the patient’s neurologic status is stable.³⁵

Intracerebral hemorrhage. Hypertension is the most important modifiable risk factor for ICH. Although long-term benefit in lowering BP in patients with ICH is widely accepted, it remains unclear if elevated BP should be lowered in the acute phase. In recent years, a few randomized trials have attempted to address the issue of acute BP lowering in ICH and are summarized in Table 41.4.^{27,36–39}

The currently recommended guidelines of the AHA/ASA and the European Stroke Initiative (ESI) for BP management in acute ICH are summarized in Table 41.5.^{40,41} The AHA/ASA guidelines also recommend avoiding peaks and large variability in SBP and initiating BP-lowering treatment within 2 hours of ICH onset and reaching target within 1 hour. If elevated ICP is of concern, such as in patients with large hemorrhages or with hydrocephalus, ICP should be monitored to ensure that the CPP is appropriate before significant BP reduction.

Subarachnoid hemorrhage. Before definitive treatment of the ruptured aneurysm, SBP is usually kept below 160 mm Hg, although there is no conclusive evidence that higher BPs increase rebleeding rates. In patients with suspected elevation of ICP, it is important to monitor ICP and keep the CPP greater than 70 mm Hg. The AHA/ASA guidelines recommend monitoring and control of BP to balance the risk for stroke, hypertension-related bleeding, and maintenance of CPP. An SBP goal of less than 160 mm Hg is considered reasonable.⁴² The ESO recommends keeping the SBP at less than 180 mm Hg but keeping the MAP at greater than 90 mm Hg.⁴³ After the ruptured aneurysm has been secured, aggressive treatment of BP should be avoided, and in the setting of cerebral vasospasm, BP is usually elevated using vasopressors until the neurologic deficits resolve, often as high as an SBP of 200 to 220 mm Hg.

Hypertension After Carotid Endarterectomy and Endovascular Procedures

Definition, Incidence, and Clinical Features

Hemodynamic disturbances such as hypotension, bradycardia, and hypertension are common (10%–40%) after carotid endarterectomy and endovascular procedures such as angioplasty and stenting. A small percentage of these patients develop carotid hyperperfusion (or reperfusion) syndrome. This syndrome occurs in the first week after surgery or angioplasty-stenting and manifests as ipsilateral pulsatile headache, seizures, ICH, transient or permanent contralateral neurologic signs, encephalopathy, or reversible cerebral edema.^{44–46} Recently, this syndrome has also been reported after treatment of acute ischemic stroke with intravenous thrombolysis and endovascular treatment. Some subjects with cerebral hyperperfusion after revascularization may not manifest clinical signs acutely but may later develop cortical neuronal loss and cognitive impairment.⁴⁷

TABLE 41.2 Major Trials of Acute Blood Pressure Reduction in Ischemic Stroke

Trial	Study Design	Study Arms	No. of Patients	Results
Acute Candesartan Cilexetil Evaluation in Stroke Survivors (ACCESS) ³²	Prospective, double-blind, placebo-controlled, randomized, multicenter, phase II	Candesartan vs. placebo	342	Lower 12-month mortality and vascular events in the candesartan group, but no significant difference in BP between the two arms.
Scandinavian Candesartan Acute Stroke Trial (SCAST) ³¹	Prospective, double-blind, placebo-controlled, randomized, multicenter	Candesartan vs. placebo	2029 (274 had cerebral hemorrhage)	During 6 months of follow-up, the risk of the composite vascular endpoint did not differ between treatment groups. Analysis of functional outcome suggested a higher risk for poor outcome in the candesartan group.
Controlling Hypertension and Hypotension Immediately Post-Stroke (CHHIPS) ²⁹	Prospective, double-blind, placebo-controlled, randomized, multicenter	Labetalol vs. lisinopril vs. placebo	179 (25 had cerebral hemorrhage)	No difference in death or dependency at 2 weeks, early neurologic deterioration, or serious adverse event.
Continue or Stop Post-stroke Antihypertensive Collaborative Study (COSSACS) ³⁰	Prospective, open, multicenter, randomized, blinded-endpoint trial	Continue vs. stop preexisting antihypertensive drugs	763 (38 had cerebral hemorrhage)	Continuation of antihypertensive drugs did not reduce 2-week death or dependency, cardiovascular event rate, or mortality at 6 months.
Glycine Antagonist in Neuroprotection (GAIN International) ²⁴	Prospective, double-blind, placebo-controlled, randomized, multicenter	Gavestinel vs. placebo	1445	A 30% drop in mean arterial pressure from baseline was not associated with poor outcome.
Intravenous Nimodipine West European Stroke Trial (INWEST) ²²	Prospective, double-blind, placebo-controlled, randomized, multicenter	Placebo vs. low-dose vs. high-dose nimodipine	265	Patients with a DBP reduction of $\geq 20\%$ in the high-dose group had a significantly increased adjusted odds ratio for the compound outcome variable of death or dependency.
Chinese Antihypertensive Trial in Acute Ischemic Stroke (CATIS) ²⁵	Prospective, single-blind, randomized, blinded endpoint, multicenter	Antihypertensive treatment vs. discontinuing all antihypertensives	4071	BP reduction with antihypertensive medications, compared with the absence of hypertensive medication, did not reduce the likelihood of death and major disability at 14 days or hospital discharge.
Efficacy of Nitric Acid in Stroke (ENOS) ²⁶	Prospective, multicenter, randomized, placebo-controlled, patient-masked, outcome-assessor-masked, parallel-group trial	Glyceryl trinitrate vs. no treatment; subset taking antihypertensive medications on admission randomized to taking vs. stopping them	4011 (629 had cerebral hemorrhage)	Transdermal glyceryl trinitrate lowered BP and had acceptable safety but did not improve functional outcome. There was no evidence to support continuing prestroke antihypertensive drugs in patients in the first few days after acute stroke.
Valsartan Efficacy on Modest Blood Pressure Reduction in Acute Ischemic Stroke (VENTURE) ²⁸	Prospective, multicenter, randomized, open-label, blinded-endpoint trial	Valsartan vs. no treatment	393	Early reduction of BP with valsartan did not reduce death or dependency and major vascular events at 90 days but increased the risk for early neurologic deterioration.
Intensive BP reduction with intravenous thrombolysis therapy for acute ischemic stroke (ENCHANTED) ²³	International, partial-factorial, open-label, blinded-endpoint	Intensive (systolic BP 130–140 mm Hg) or guideline (systolic BP < 180 mm Hg) antihypertensive treatment for 72 hours in patients within 6 hours of stroke onset and treated with intravenous alteplase	2227	Fewer patients in the intensive group had intracranial hemorrhage, but there was no difference in clinical outcome between the two groups.
Prehospital transdermal glyceryl trinitrate in patients with ultra-acute presumed stroke (RIGHT-2) ²⁷	Prospective, multicentric, paramedic-delivered, ambulance-based, randomized, sham-controlled, blinded endpoint	Transdermal nitroglycerin vs. sham dressing	1149 (597 with ischemic stroke, 145 with cerebral hemorrhage, 109 with transient ischemic attack, 297 with a nonstroke mimic)	Nitroglycerin did not lead to an improved outcome at 90 days.

BP, Blood pressure; DBP, diastolic blood pressure.

TABLE 41.3 Guidelines for Blood Pressure Management After Acute Ischemic Stroke

AHA/ASA	ESO
Patients Eligible for Thrombolytic Therapy or Mechanical Thrombectomy Before Thrombolytic Therapy Lower SBP to <185 and DBP to <110 mm Hg.	- Routine BP lowering is not recommended. - Cautious BP lowering is recommended in patients with extremely high BP (>220/120 mm Hg) on repeated measurements, or with severe cardiac failure, aortic dissection, or hypertensive encephalopathy.
During and After Thrombolytic Therapy or Mechanical Thrombectomy Lower SBP to <180 and DBP to <105 mm Hg.	- Abrupt BP lowering should be avoided. - BP must be <185/110 mm Hg before, and for the first 24 hours after, thrombolysis.
Patients Not Eligible for Thrombolytic Therapy - Patients with markedly elevated BP may have their BP lowered. - Lowering BP by ~15% in these patients is reasonable. - Antihypertensive drugs should be withheld unless SBP >220 or DBP >120 mm Hg.	

AHA/ASA, American Heart Association/American Stroke Association; BP, blood pressure; DBP, diastolic blood pressure; ESO, European Stroke Organisation; SBP, systolic blood pressure.

After carotid endarterectomy, the incidence of postoperative severe hypertension is reported as 19% and that of carotid hyperperfusion syndrome as 1%. Most cases occurred in the first week and the average time to symptoms was the fifth postoperative day. Seizures (36%), hemiparesis (31%), or both (33%) were common presenting features, and 59% of patients had headache.⁴⁸

Pathophysiology

Preexisting hypertension, baroreceptor impairment after surgical manipulation, and elevated catecholamine levels after cerebral hypoperfusion during intraoperative cross-clamping may contribute to postoperative hypertension that contributes to cerebral hyperperfusion, which may in turn lead to cerebral hemorrhage. The hyperperfusion syndrome may be due, in part, to impaired autoregulation from chronic vasodilation of the distal vascular bed ipsilateral to a hemodynamically significant internal carotid artery stenosis.⁴⁹ Other postulated mechanisms include activation of the trigeminovascular axon reflex and derangement of the carotid baroreceptors.⁵⁰ Subjects at risk for development of this syndrome are those with extensive microvascular disease, preoperative hypoperfusion and impaired autoregulation, or postoperative hyperperfusion.

Diagnosis and Treatment

Cerebral hyperperfusion represents a postoperative increase in CBF of more than 100% compared with preoperative flow. However, this increase in blood flow may be only approximately 20% compared with the contralateral side.⁵¹ Making a diagnosis based on CBF doubling alone may lead to a significant overestimation of the incidence, and the following four criteria have been suggested: (1) occurrence within 30

TABLE 41.4 Major Trials of Acute Blood Pressure Reduction in Cerebral Hemorrhage

Trial	Study Design	Study Arms	No. of Patients	Results
Koch et al. ³⁸	Prospective, randomized	Target MBP <110 or 110–130 mm Hg	42	No significant differences in early neurologic deterioration, hematoma and edema growth, and clinical outcome.
INTERACT ³⁶	Prospective, randomized, blinded endpoint assessment	Target SBP <140 or <180 mm Hg	404	No significant difference in hematoma growth between the two tiers after adjustment for baseline hematoma volume and time to CT scan.
INTERACT 2 ⁶³	Prospective, randomized, blinded endpoint assessment	Target SBP <140 or <180 mm Hg	2839	No significant reduction in the rate of primary outcome of death or severe disability with intensive lowering of BP. Improved functional outcomes with lower BP target on ordinal analysis of modified Rankin Scale score.
ATACH ³⁷	Prospective, Phase I dose-escalation	Target SBP 110–140 or 140–170 or 170–200 mm Hg	60	Observed proportions of neurologic deterioration and serious adverse events below the prespecified safety thresholds; 3-month mortality rate lower than expected in all tiers.
ATACH 2 ³⁹	Prospective, randomized, multicenter, open-label trial	Target SBP 110–140 or 140–179 mm Hg	1000	BP reduction to a target SBP of 110–139 mm Hg did not result in a lower rate of death or disability than a target of 140–179 mm Hg. The rate of renal adverse events within 7 days after randomization was significantly higher in the group with the lower BP target.
Prehospital transdermal glyceryl trinitrate in patients with ultra-acute presumed stroke (RIGHT-2) ²⁷	Prospective, multicentric, paramedic-delivered, ambulance-based, randomized, sham-controlled, blinded end-point	Transdermal nitroglycerin versus sham dressing	1149 (597 with ischemic stroke, 145 with cerebral hemorrhage, 109 with transient ischemic attack, 297 with a nonstroke mimic)	Nitroglycerin did not lead to an improved outcome at 90 days.

BP, Blood pressure; CT, computed tomography; MAP, mean arterial blood pressure; SBP, systolic BP.

TABLE 41.5 Guidelines for Blood Pressure Management After Acute Cerebral Hemorrhage

AHA/ASA	ESO
For patients with mild to moderate severity ICH presenting with SBP between 150 and 200 mm Hg, acute lowering of SBP to a goal of 140 mm Hg is safe and may be reasonable for improving functional outcomes (Class 2b; Level of Evidence B-R). In patients with large or severe ICH and in those requiring surgical decompression, the safety and efficacy of intensive BP lowering are not well established (Class 2b, Level of Evidence C-LD). In patients with mild to moderate severity ICH presenting with SBP >150 mm Hg, acute lowering of SBP to <130 mm Hg is probably harmful (Class 3, Level of Evidence B-R).	In acute ICH within 6 hours of onset, intensive BP reduction (SBP target <140 mm Hg in <1 hour) is safe and may be superior to SBP target of <180 mm Hg. No specific agent can be recommended. (quality of evidence: moderate; strength of recommendation: weak).

AHA/ASA, American Heart Association/American Stroke Association; BP, blood pressure; ESO, European Stroke Organisation; ICH, intracerebral hemorrhage; SBP, systolic BP.

days post-carotid endarterectomy; (2) evidence of hyperperfusion (on transcranial Doppler, single-photon emission computed tomography, or computed tomography [CT]/magnetic resonance [MR] perfusion imaging) or SBP greater than 180 mm Hg; (3) clinical features such as new headache, seizures, hemiparesis, and Glasgow Coma Scale score less than 15 or radiologic features such as cerebral edema or ICH; and (4) no evidence of new cerebral ischemia, postoperative carotid occlusion, or metabolic or pharmacologic cause to explain the findings.⁴⁸

Because of the risk for development of carotid hyperperfusion syndrome after carotid endarterectomy or stenting, all patients should have continuous intraoperative and postoperative BP monitoring. Most authors advocate strict BP control (SBP <120 mm Hg) from the time of intraoperative internal carotid artery unclamping or angioplasty, particularly in high-risk patients.⁵² If high-risk features are absent, aiming for SBP of 140 to 160 mm Hg or preoperative SBP (if lower) in the postoperative period is reasonable. Elevated BP should be treated with intravenous (IV) labetalol or clonidine. Vasodilators such as nitroglycerin, hydralazine, and sodium nitroprusside should be avoided. Because this syndrome may occur after patients have been discharged from the hospital, it is important for physicians to ensure that BP is appropriately controlled even after the immediate postoperative period.

Hypertension After Spinal Cord Injury

Definition and Epidemiology

Hypertension is quite prevalent in patients with spinal cord injury (SCI), with a reported prevalence of 14% to 61%. The prevalence is higher in patients with paraplegia compared with tetraplegia and in those with nontraumatic SCI compared with those with traumatic SCI.⁵³

In addition, autonomic dysreflexia occurs in up to 70% of persons after SCI, most often during the first 2 to 4 months after injury. It is defined as an increase in SBP by at least 20%, associated with a change in heart rate, and accompanied by at least one sign (sweating, piloerection, facial flushing) or symptom (headache, blurred vision, stuffy nose).⁵⁴ If autonomic dysreflexia is unrecognized, it can result

in serious sequelae, such as posterior leukoencephalopathy, ICH, SAH, seizures, arrhythmia, pulmonary edema, retinal hemorrhage, and, rarely, coma or death.⁵⁵

Pathophysiology and Diagnosis of Autonomic Dysreflexia

Autonomic dysreflexia is most commonly seen in patients with complete SCI in which there is loss of all neurologic function below the level of the lesion. The spinal cord lesion is typically at or above the sixth thoracic spinal level. Immediately after the injury, there is initial loss of supraspinal sympathetic control similar to the initial period of muscle flaccidity. This often leads to hypotension and bradycardia (spinal shock). After a few weeks to months, there is extrajunctional sprouting of the α -receptors, denervation hypersensitivity, and impaired presynaptic uptake of norepinephrine. In addition, there may be derangement of spinal glutaminergic interneurons. Noxious stimuli below the neurologic level of the lesion trigger a spinal reflex arc that results in increased sympathetic tone and hypertension.⁵⁶ The most common inciting events are an overdistended urinary bladder and fecal impaction. However, it may be secondary to other precipitants, including infections, pressure ulcers, urologic and endoscopic procedures, sympathomimetic medications, and sildenafil citrate used for sperm retrieval.⁵⁷

Clinical symptoms include pulsatile headache, blurred vision, anxiety, nasal congestion, nausea, and sweating above the involved spinal level. The flushed, sweaty skin above the lesion level is because of brainstem parasympathetic activation. At and below the lesion, the skin remains pale, cool, and dry. Heart rate can be quite variable from bradycardia to tachycardia. The hallmark physical finding is elevated BP. However, because BP may normally be quite low after SCI, baseline BP readings may be within the normal range but elevated for a given individual, making clinical suspicion and reliance on other clinical signs and symptoms paramount in the diagnosis if baseline BP is not known.⁵⁶

Treatment

Although it is recognized that BP is often lower in patients with tetraplegia and high paraplegia, current guidelines suggest applying evidence-based guidelines for treating hypertension in the general population to SCI patients as well. However, SCI may affect the choice of the antihypertensive agent. Thiazide diuretics are not preferred in patients who perform intermittent bladder catheterization because they increase urine volume. In addition, SCI patients may experience electrolyte abnormalities or decline in renal function during thiazide use. Some patients with SCI may experience supine hypertension with coexisting orthostatic hypotension, which requires cautious trial and titration of antihypertensive agents.⁵³

Vigilant preventive measures for autonomic dysreflexia include proper bowel, bladder, and skin care. However, expeditious treatment of elevated BP is critical to avoid the potentially life-threatening consequences. Placement of the patient upright with the legs lowered to reduce BP and removal of any possible noxious stimuli (such as binding clothing and devices) are the initial treatments. It is also important to look for and appropriately treat other common triggers such as urinary retention or constipation.

Pharmacologic treatment with rapid-acting, short-lived agents may be indicated for SBP elevation of 150 mm Hg or greater that persists after the preceding interventions. Nitroglycerin is often used to treat hypertension associated with autonomic dysreflexia. However, to avoid precipitating hypotension, nitrate-containing agents should not be given for 24 hours before the use of sildenafil or similar agents to facilitate sperm retrieval or treat erectile dysfunction in SCI. CCBs and ACE inhibitors

TABLE 41.6 Preferred Antihypertensive Agents in the Treatment of Stroke-Associated Hypertension

Drug	Mechanism of Action	Intravenous Dose	Advantages	Disadvantages
Labetalol ^a	α_1 -, β_1 -, and β_2 -receptor antagonist	10 mg IV over 1–2 min, may repeat once; IV infusion 2–8 mg/min	Does not lower CBF Does not increase ICP	May exacerbate bradycardia
Nicardipine ^a	L-type CCB	5–15 mg/hr	Does not decrease CBF	May increase ICP Long duration of action
Clevidipine ^a	L-type CCB	Starting dose 1–2 mg/hr; dose can be doubled every 2–5 min up to a maximum dose of 21 mg/hr	Does not decrease CBF	Should be avoided in patients with egg/soy allergy
Esmolol	β_1 -receptor antagonist	500- μ g/kg bolus, then 50–300 μ g/kg/min	Does not lower CBF Does not increase ICP	May exacerbate bradycardia
Sodium nitroprusside	Vasodilator	0.25–10 μ g/kg/min	Potent antihypertensive	May increase ICP Can cause cerebral steal Potential for cyanide toxicity
Nitroglycerin	Vasodilator	5–200 μ g/min	Can be helpful for concomitant cardiac ischemia	May increase ICP Can cause cerebral steal
Hydralazine	Vasodilator	2.5- to 10-mg bolus	Can be given as IV bolus when labetalol is contraindicated because of bradycardia	May increase ICP Can cause cerebral steal
Enalaprilat	ACE inhibitor	0.625–1.25 mg every 6 hr	Does not decrease CBF	Variable response Long duration of action

^aPreferred agents listed in 2018 American Heart Association/American Stroke Association guidelines.

ACE, Angiotensin-converting enzyme; CBF, cerebral blood flow; CCB, calcium channel blocker; ICP, intracranial pressure; IV, intravenous.

also have been reported to be effective.⁵⁸ However, these agents might exacerbate resting hypotension and therefore should be used with caution. Prazosin has been reported to be effective in reducing the severity of autonomic dysreflexia without significantly lowering the resting BP. If the bladder is empty and the BP is less than 150 mm Hg, fecal disimpaction with topical anesthetic should be attempted. If dysreflexia is refractory or associated with severe clinical presentation, other precipitants should be sought, and hospitalization may be indicated.⁵⁹

Up to 90% of pregnant women with upper SCI experience autonomic dysreflexia during labor and delivery. Appropriate epidural or spinal anesthesia techniques can ameliorate the risk.⁶⁰

Cerebrovascular Effects of Antihypertensive Agents

Different classes of antihypertensive agents have different direct effects on the CBF, ICP, and autoregulation. The ideal drug would not increase ICP or decrease blood flow to ischemic regions. In addition, in treatment of hypertension in the acute setting, drugs that can be given by IV, have a short half-life, and do not cause sedation are preferable. In the chronic phase after a stroke, adequate BP lowering is key, and there is no clear evidence favoring one class of antihypertensive agent over another; however, one may consider avoiding the use of nonselective β -blockers that may increase SBP variability.

The advantages and disadvantages of various classes of antihypertensive agents after acute stroke are summarized in Table 41.6.

β -Adrenergic antagonists (e.g., esmolol) and combined α - and β -adrenergic receptor antagonists (e.g., labetalol) do not increase ICP or affect cerebral autoregulation. They are suitable for treatment of hypertension in the setting of acute cerebral ischemia or increased ICP. However, bradycardia secondary to increased ICP is a relative contraindication.

Vasodilators (e.g., sodium nitroprusside, nitroglycerin) cause cerebral arterial dilation and venodilation and can theoretically increase ICP and cause a cerebral steal phenomenon in patients with acute cerebral ischemia. Other disadvantages of sodium nitroprusside are

tachyphylaxis, the need to shield the medication from light because of its photosensitivity, and the danger of cyanide and thiocyanate toxicity that may be difficult to detect in patients with brain injury. However, they may be used in patients with small and moderate-sized ICH and in patients with SAH if increased ICP is not a concern.

CCBs have varying effects on cerebral autoregulation. Nifedipine can lead to severe reduction in BP and is not recommended. Nimodipine is used routinely in patients with SAH because it has been shown to improve outcome, possibly because of a neuroprotective effect. Nicardipine has been used in patients with acute ICH without any change in CBF and is often used in patients with SAH. It is becoming quite popular in neurocritical care units because of its efficacy, ease of titration, predictable response, and favorable cerebral hemodynamic effects. Clevidipine has been used to lower BP in patients with ischemic and hemorrhagic stroke with a similar safety and efficacy profile as nicardipine.

ACE inhibitors and the ARB candesartan have been used in patients with acute cerebral ischemia and have no effect on CBF. However, short-acting parenteral forms of these drugs are not available. ACE inhibitors and ARBs shift the lower limit of cerebral autoregulation toward lower BP in rats and humans. However, these agents have a long half-life, which is not desirable in treatment of hypertension in the acute phase.

Similarly, because of its long half-life and sedative effect, the α_2 -adrenergic agonist clonidine is not preferred.

The cerebrovascular effects of the newer parenteral antihypertensives such as fenoldopam, a peripheral dopamine-1 receptor agonist, and clevidipine, a CCB, have not been extensively studied. However, in small studies, fenoldopam has been shown to decrease global CBF and increase ICP in patients with impaired intracranial compliance.⁶¹ In a small single-center study, clevidipine was found to be safe and effective in perioperative neurosurgical patients with hypertension.⁶²

In practice, we prefer IV boluses of labetalol and hydralazine to treat elevated BP in the setting of an acute stroke. If frequent doses are required, or the BP is extremely elevated, an IV infusion of nicardipine or clevidipine is used.

SUMMARY

This chapter provides a summary of neurologic pathways that may be responsible for mediating systemic BP control and emphasizes management of BP in acute ischemic and hemorrhagic stroke subtypes.

SELF-ASSESSMENT QUESTIONS

1. A 55-year-old man with longstanding hypertension presents with acute right hemiplegia. The onset of symptoms was 45 minutes before presentation. His BP on presentation is 200/110 mm Hg and his noncontrast head CT scan is normal. He is given 10 mg labetalol and his BP is now 175/105 mm Hg. IV tissue plasminogen activator (tPA) is administered for acute ischemic stroke. An hour later, his BP is 170/110 mm Hg. Which of the following statements *best* characterizes BP management in this case?
 - A. The patient should not have been given IV tPA because his BP (175/105 mm Hg) was too high.
 - B. The patient's current BP of 170/110 mm Hg needs to be lowered with IV labetalol or nicardipine.
 - C. Sublingual nifedipine is the drug of choice in the acute management of elevated BP in the setting of an acute stroke.
 - D. BP should be increased using vasopressors to augment cerebral perfusion.
 - E. None of the above.
2. A 60-year-old man with poorly controlled BP presents to the emergency department with left hemiparesis that was first noticed 2 days ago. He is awake, fully alert, and oriented. His strength on the right side is normal. His head CT scan shows a 2×2 cm right putaminal hemorrhage with no hydrocephalus or intraventricular extension. His BP is 150/85 mm Hg. Which of the following statements *best* characterizes BP management in this case?
 - A. This patient has a greater than 50% chance of clinically significant expansion of his ICH over the next 48 hours if his BP is not lowered acutely.
 - B. BP should be lowered immediately to less than 120/80 mm Hg with intravenous nicardipine to prevent hematoma expansion.
 - C. He should have an intracranial pressure monitor inserted to measure intracranial pressure.
 - D. There is an area of ischemia surrounding an ICH that may increase in size with additional BP lowering.
 - E. None of the above.
3. Which of the following features *best* characterizes carotid hyperperfusion syndrome after a carotid endarterectomy?
 - A. Headache
 - B. Intracerebral hemorrhage
 - C. Increase in cerebral blood flow ipsilateral to the endarterectomy
 - D. Seizures
 - E. All of the above

REFERENCES

- Qureshi AI. Acute hypertensive response in patients with stroke: pathophysiology and management. *Circulation*. 2008;118:176–187.
- Parati G, Esler M. The human sympathetic nervous system: its relevance in hypertension and heart failure. *Euro Heart J*. 2012;33:1058–1066.
- Chalmers J. Volhard lecture. Brain, blood pressure and stroke. *J Hypertens*. 1998;16:1849–1858.
- Colombari E, Sato MA, Cravo SL, Bergamaschi CT, Campos Jr RR, Lopes OU. Role of the medulla oblongata in hypertension. *Hypertension*. 2001;38:549–554.
- Talman WT. Cardiovascular regulation and lesions of the central nervous system. *Ann Neurol*. 1985;18:1–13.
- Fisher JP, Paton JF. The sympathetic nervous system and blood pressure in humans: implications for hypertension. *J Hum Hypertens*. 2012;26:463–475.
- Strandgaard S, Olesen J, Skinhoj E, Lassen NA. Autoregulation of brain circulation in severe arterial hypertension. *Br Med J*. 1973;1:507–510.
- Strandgaard S. Autoregulation of cerebral blood flow in hypertensive patients. The modifying influence of prolonged antihypertensive treatment on the tolerance to acute, drug-induced hypotension. *Circulation*. 1976;53:720–727.
- Mann SJ. Neurogenic hypertension: pathophysiology, diagnosis and management. *Clin Auton Res*. 2018;28:363–374.
- Qureshi AI, Ezzeddine MA, Nasar A, et al. Prevalence of elevated blood pressure in 563,704 adult patients with stroke presenting to the ED in the United States. *Am J Emerg Med*. 2007;25:32–38.
- CAST: randomised placebo-controlled trial of early aspirin use in 20,000 patients with acute ischaemic stroke. CAST (Chinese Acute Stroke Trial) Collaborative Group. *Lancet*. 1997;349:1641–1649.
- The International Stroke Trial (IST): a randomised trial of aspirin, subcutaneous heparin, both, or neither among 19435 patients with acute ischaemic stroke. International Stroke Trial Collaborative Group. *Lancet*. 1997;349:1569–1581.
- Pedely L, Gorelick PB. Chronic management of blood pressure after stroke. *Hypertension*. 2004;44:1–5.
- Lawes CM, Bennett DA, Feigin VL, Rodgers A. Blood pressure and stroke: an overview of published reviews. *Stroke*. 2004;35:776–785.
- MacMahon S, Peto R, Cutler J, et al. Blood pressure, stroke, and coronary heart disease. Part 1, prolonged differences in blood pressure: prospective observational studies corrected for the regression dilution bias. *Lancet*. 1990;335:765–774.
- Group PC. Randomised trial of a perindopril-based blood-pressure-lowering regimen among 6,105 individuals with previous stroke or transient ischaemic attack. *Lancet*. 2001;358:1033–1041.
- Aiyagari V, Gorelick PB. Management of blood pressure for acute and recurrent stroke. *Stroke*. 2009;40:2251–2256.
- Testai FD, Aiyagari V. Acute hemorrhagic stroke pathophysiology and medical interventions: blood pressure control, management of anticoagulant-associated brain hemorrhage and general management principles. *Neurol Clin*. 2008;26:963–985.
- Davis SM, Broderick J, Hennerici M, et al. Hematoma growth is a determinant of mortality and poor outcome after intracerebral hemorrhage. *Neurology*. 2006;66:1175–1181.
- Butcher KS, Jeerakathil T, Hill M, et al. The intracerebral hemorrhage acutely decreasing arterial pressure trial. *Stroke*. 2013;44:620–626.
- Geeganage C, Bath PM. Interventions for deliberately altering blood pressure in acute stroke. *Cochrane Database Syst Rev*. 2008;CD000039.
- Ahmed N, Nasman P, Wahlgren NG. Effect of intravenous nimodipine on blood pressure and outcome after acute stroke. *Stroke*. 2000;31:1250–1255.
- Anderson CS, Huang Y, Lindley RI, et al. Intensive blood pressure reduction with intravenous thrombolysis therapy for acute ischaemic stroke (ENCHANTED): an international, randomised, open-label, blinded-endpoint, phase 3 trial. *Lancet*. 2019;393:877–888.
- Aslanyan S, Fazekas F, Weir CJ, et al. Effect of blood pressure during the acute period of ischemic stroke on stroke outcome: a tertiary analysis of the GAIN international trial. *Stroke*. 2003;34:2420–2425.
- He J, Zhang Y, Xu T, et al. Effects of immediate blood pressure reduction on death and major disability in patients with acute ischemic stroke: the CATIS randomized clinical trial. *J Am Med Assoc*. 2014;311:479–489.
- Investigators ET. Efficacy of nitric oxide, with or without continuing antihypertensive treatment, for management of high blood pressure in acute stroke (ENOS): a partial-factorial randomised controlled trial. *Lancet*. 2015;385:617–628.
- Investigators R-. Prehospital transdermal glyceryl trinitrate in patients with ultra-acute presumed stroke (RIGHT-2): an ambulance-based, randomised, sham-controlled, blinded, phase 3 trial. *Lancet*. 2019;393:1009–1020.
- Oh MS, Yu KH, Hong KS, et al. Modest blood pressure reduction with valsartan in acute ischemic stroke: a prospective, randomized, open-label, blinded-end-point trial. *Int J Stroke*. 2015;10:745–751.
- Potter JF, Robinson TG, Ford GA, et al. Controlling hypertension and hypotension immediately post-stroke (CHHIPS): a randomised, placebo-controlled, double-blind pilot trial. *Lancet Neurol*. 2009;8:48–56.
- Robinson TG, Potter JF, Ford GA, et al. Effects of antihypertensive treatment after acute stroke in the Continue or Stop Post-Stroke Antihypertensives Collaborative Study (COSSACS): a prospective, randomised, open, blinded-endpoint trial. *Lancet Neurol*. 2010;9:767–775.
- Sandset EC, Bath PM, Boysen G, et al. The angiotensin-receptor blocker candesartan for treatment of acute stroke (SCAST): a randomised, placebo-controlled, double-blind trial. *Lancet*. 2011;377:741–750.
- Schrader J, Luders S, Kulschewski A, et al. The ACCESS study: evaluation of acute candesartan cilexetil therapy in stroke survivors. *Stroke*. 2003;34:1699–1703.
- Ahmed N, Audebert H, Turc G, et al. Consensus statements and recommendations from the ESO-karolinska stroke update conference, Stockholm 11–13 November 2018. *Eur Stroke J*. 2019;4:307–317.
- Powers WJ, Rabinstein AA, Ackerson T, et al. 2018 Guidelines for the early management of patients with acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2018;49:e46–e110.
- Gorelick PB, Whelton PK, Sorond F, Carey RM. Blood pressure management in stroke. *Hypertension*. 2020;76:1688–1695.
- Anderson CS, Huang Y, Wang JG, et al. Intensive blood pressure reduction in acute cerebral haemorrhage trial (INTERACT): a randomised pilot trial. *Lancet Neurol*. 2008;7:391–399.
- Antihypertensive Treatment of Acute Cerebral Hemorrhage. Antihypertensive treatment of acute cerebral hemorrhage. *Critical Care Med*. 2010;38:637–648.
- Koch S, Romano JG, Forteza AM, Otero CM, Rabinstein AA. Rapid blood pressure reduction in acute intracerebral hemorrhage: feasibility and safety. *Neurocrit Care*. 2008;8:316–321.
- Qureshi AI, Palesch YY, Barsan WG, et al. Intensive blood-pressure lowering in patients with acute cerebral hemorrhage. *New Engl J Med*. 2016;375:1033–1043.
- Greenberg SM, Ziai WC, Cordonnier C, et al. 2022 guideline for the management of patients with spontaneous intracerebral hemorrhage: a guideline from the American Heart Association/American Stroke Association. *Stroke*. 2022;53:e282–e361.
- Steiner T, Al-Shahi Salman R, Beer R, et al. European Stroke Organisation (ESO) guidelines for the management of spontaneous intracerebral hemorrhage. *Int J Stroke*. 2014;9:840–855.
- Connolly Jr ES, Rabinstein AA, Carhuapoma JR, et al. Guidelines for the management of aneurysmal subarachnoid hemorrhage: a guideline for healthcare professionals from the American heart association/American stroke association. *Stroke*. 2012;43:1711–1737.
- Steiner T, Juvela S, Unterberg A, et al. European stroke organization guidelines for the management of intracranial aneurysms and subarachnoid haemorrhage. *Cerebrovasc Dis*. 2013;35:93–112.
- Qureshi AI, Luft AR, Sharma M, et al. Frequency and determinants of postprocedural hemodynamic instability after carotid angioplasty and stenting. *Stroke*. 1999;30:2086–2093.
- Wade JG, Larson Jr CP, Hickey RF, Ehrenfeld WK, Severinghaus JW. Effect of carotid endarterectomy on carotid chemoreceptor and baroreceptor function in man. *N Engl J Med*. 1970;282:823–829.

46. Wong JH, Findlay JM, Suarez-Almazor ME. Hemodynamic instability after carotid endarterectomy: risk factors and associations with operative complications. *Neurosurgery*. 1997;41:35–41.
47. Chida K, Ogasawara K, Suga Y, et al. Postoperative cortical neural loss associated with cerebral hyperperfusion and cognitive impairment after carotid endarterectomy: 123I-iomazenil SPECT study. *Stroke*. 2009;40:448–453.
48. Bourri S, Thapar A, Shalhoub J, et al. Hypertension and the post-carotid endarterectomy cerebral hyperperfusion syndrome. *Eur J Vasc Endovasc Surg*. 2011;41:229–237.
49. Yoshimoto T, Houkin K, Kuroda S, Abe H, Kashiwaba T. Low cerebral blood flow and perfusion reserve induce hyperperfusion after surgical revascularization: case reports and analysis of cerebral hemodynamics. *Surg Neurol*. 1997;48:132–138.
50. van Mook WN, Rennenberg RJ, Schurink GW, et al. Cerebral hyperperfusion syndrome. *Lancet Neurol*. 2005;4:877–888.
51. Karapanayiotides T, Meuli R, Devuyst G, et al. Postcarotid endarterectomy hyperperfusion or reperfusion syndrome. *Stroke*. 2005;36:21–26.
52. Abou-Chebl A, Reginelli J, Bajzer CT, Yadav JS. Intensive treatment of hypertension decreases the risk of hyperperfusion and intracerebral hemorrhage following carotid artery stenting. *Catheter Cardiovasc Interv*. 2007;69:690–696.
53. Nash MS, Groah SL, Gater Jr DR, et al. Identification and management of cardiometabolic risk after spinal cord injury: clinical practice guideline for health care providers. *Top Spinal Cord Inj Rehabil*. 2018;24:379–423.
54. Furlan JC, Fehlings MG. Cardiovascular complications after acute spinal cord injury: pathophysiology, diagnosis, and management. *Neurosurg Focus*. 2008;25:E13.
55. Valles M, Benito J, Portell E, Vidal J. Cerebral hemorrhage due to autonomic dysreflexia in a spinal cord injury patient. *Spinal Cord*. 2005;43:738–740.
56. Blackmer J. Rehabilitation medicine: 1. Autonomic dysreflexia. *CMAJ*. 2003;169:931–935.
57. Sheel AW, Krassioukov AV, Inglis JT, Elliott SL. Autonomic dysreflexia during sperm retrieval in spinal cord injury: influence of lesion level and sildenafil citrate. *J Appl Physiol (1985)*. 2005;99:53–58.
58. Hagen EM, Rekand T, Gronning M, Faerestrland S. Cardiovascular complications of spinal cord injury. *Tidsskr Nor Lægeforen*. 2012;132:1115–1120.
59. Consortium for Spinal Cord M. Acute management of autonomic dysreflexia: individuals with spinal cord injury presenting to health-care facilities. *J Spinal Cord Med*. 2002;25(suppl 1):S67–S88.
60. Ribes Pastor MP, Vanarase M. Peripartum anaesthetic management of a parturient with spinal cord injury and autonomic hyperreflexia. *Anaesthesia*. 2004;59:94.
61. Rhoney DH, Liu-DeRyke X. Effect of vasoactive therapy on cerebral circulation. *Critical care clinics*. 2006;22:221–243, vi.
62. Bekker A, Didehvar S, Kim S, et al. Efficacy of clevidipine in controlling perioperative hypertension in neurosurgical patients: initial single-center experience. *J Neurosurg Anesthesiol*. 2010;22:330–335.
63. Anderson CS, Heeley E, Huang Y, et al. Rapid blood-pressure lowering in patients with acute intracerebral hemorrhage. *N Engl J Med*. 2013;368:2355–2365.